RATIONAL BRIDGES STAY RATIONAL BRIDGES AFTER BASE FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument:

Editorial note. The label lemma-rational-bridge-field-extension names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The rational bridge setup. Same chapter, Example \label{example-rational-bridge}. Proof lines 6344-6345: k"/k'/k are the field extensions of the example; k'/k is finite separable and [k" : k'] = 2.
- Cohomology under flat base change. Cohomology of Schemes, Lemma \label{coherent-lemma-flat-base-change-cohomology}. Proof lines 6354-6356: H^1(C_K, O_{C_K}) = 0 by flat base change, so each C_i has genus 0 over K'_i.

Source reference. Original reference 0E3M: sources/stacks-curves/pinned-original-source.tex, source lines 6201–6201. Prerequisite pointer invoked at proof line(s) 6345; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Assume the singularities of X are at-worst-nodal. Let $C \subset X$ be a rational bridge (Example 0E3M). For any field extension K/k the base change $C_K \subset X_K$ is a finite disjoint union of rational bridges.*

Proof. Let $k''/k'/k$ be as in the example. Since k'/k is finite separable, we see that $k' \otimes_k K = K'_1 \times \ldots \times K'_n$ is a finite product of finite separable extensions K'_i/K. The corresponding product decomposition $k'' \otimes_k K = \prod K''_i$ gives degree 2 separable algebra extensions K''_i/K'_i. Set $C_i = C_{K'_i}$. Then $C_K = \coprod C_i$ and therefore each C_i has genus 0 (viewed as a curve over K'_i), because $H^1(C_K, \mathcal{O}_{C_K}) = 0$ by flat base change. Finally, we have $X'_K \cap C_i = \mathrm{Spec}(K''_i)$ has degree 2 over K'_i as desired. □

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